

**NOTE: This file is still under construction and may contain errors.**

The necessary input data are summarized in the tables below, which show:

- a) the values of  $(n, k, |Z|, \text{adim}, \text{vdim})$ ,
- b) the exponents  $(a_1, \dots, a_N)$ ,
- c) the coordinates and multiplicities of the points in the configuration. Note that the syzygy-based construction may increase the multiplicity at a given point.

| Example 1                               |                     |                 |                 |
|-----------------------------------------|---------------------|-----------------|-----------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 22, 1, -3)$ |                 |                 |
| $(a_1, \dots, a_N)$                     | $(5, 7, 9)$         |                 |                 |
| (coordinates); multiplicity             | (0, 0, 0, 1); 1     | (0, 0, 1, 0); 1 | (0, 0, 1, 1); 1 |
|                                         | (0, 0, 1, 2); 1     | (0, 1, 0, 0); 1 | (0, 1, 0, 1); 1 |
|                                         | (0, 1, 0, 2); 1     | (0, 1, 1, 1); 1 | (0, 1, 1, 2); 1 |
|                                         | (0, 1, 1, 3); 1     | (1, 0, 0, 0); 1 | (1, 0, 1, 0); 1 |
|                                         | (1, 0, 1, 1); 1     | (1, 0, 1, 2); 1 | (1, 0, 2, 2); 1 |
|                                         | (1, 1, 1, 1); 1     | (1, 1, 1, 2); 1 | (1, 1, 1, 3); 1 |
|                                         | (1, 1, 2, 2); 1     | (1, 1, 2, 3); 1 | (1, 1, 2, 4); 1 |
|                                         |                     | (1, 2, 2, 4); 1 |                 |

| Example 2                               |                     |                 |                 |
|-----------------------------------------|---------------------|-----------------|-----------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 24, 1, -5)$ |                 |                 |
| $(a_1, \dots, a_N)$                     | $(5, 7, 11)$        |                 |                 |
| (coordinates); multiplicity             | (0, 0, 0, 1); 1     | (0, 0, 1, 0); 1 | (0, 0, 1, 1); 1 |
|                                         | (0, 0, 1, 2); 1     | (0, 1, 0, 0); 1 | (0, 1, 0, 1); 1 |
|                                         | (0, 1, 1, 1); 1     | (0, 1, 1, 2); 1 | (0, 2, 1, 2); 1 |
|                                         | (1, 0, 0, 0); 1     | (1, 0, 1, 0); 1 | (1, 0, 1, 1); 1 |
|                                         | (1, 0, 1, 2); 1     | (1, 0, 2, 2); 1 | (1, 1, 1, 1); 1 |
|                                         | (1, 1, 1, 2); 1     | (1, 1, 2, 2); 1 | (1, 1, 2, 3); 1 |
|                                         | (1, 2, 1, 2); 1     | (1, 2, 2, 2); 1 | (1, 2, 2, 3); 1 |
|                                         | (1, 2, 2, 4); 1     | (1, 2, 3, 4); 1 | (2, 2, 3, 4); 1 |

| Example 3                               |                                                                                                                                                                      |                                                                                                                                                                      |                                                                                                                                                                                           |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 25, 1, 0)$                                                                                                                                                   |                                                                                                                                                                      |                                                                                                                                                                                           |
| $(a_1, \dots, a_N)$                     | $(7, 8, 9)$                                                                                                                                                          |                                                                                                                                                                      |                                                                                                                                                                                           |
| (coordinates); multiplicity             | $(0, 0, 0, 1); 1$<br>$(0, 0, 1, 2); 1$<br>$(0, 1, 1, 0); 1$<br>$(0, 1, 2, 2); 1$<br>$(1, 1, 0, 1); 1$<br>$(1, 1, 1, 2); 1$<br>$(1, 2, 1, 1); 1$<br>$(1, 2, 2, 2); 1$ | $(0, 0, 1, 0); 1$<br>$(0, 1, 0, 0); 1$<br>$(0, 1, 1, 1); 1$<br>$(1, 0, 0, 0); 1$<br>$(1, 1, 1, 0); 1$<br>$(1, 1, 2, 2); 1$<br>$(1, 2, 1, 2); 1$<br>$(1, 2, 2, 3); 1$ | $(0, 0, 1, 1); 1$<br>$(0, 1, 0, 1); 1$<br>$(0, 1, 1, 2); 1$<br>$(1, 1, 0, 0); 1$<br>$(1, 1, 1, 1); 1$<br>$(1, 2, 0, 1); 1$<br>$(1, 2, 1, 3); 1$<br>$(1, 3, 2, 3); 1$<br>$(2, 3, 2, 3); 1$ |

| Example 4                               |                                                                                                                                                                                           |                                                                                                                                                                                           |                                                                                                                                                                                                                |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 28, 1, -3)$                                                                                                                                                                       |                                                                                                                                                                                           |                                                                                                                                                                                                                |
| $(a_1, \dots, a_N)$                     | $(7, 9, 11)$                                                                                                                                                                              |                                                                                                                                                                                           |                                                                                                                                                                                                                |
| (coordinates); multiplicity             | $(0, 0, 0, 1); 1$<br>$(0, 0, 1, 2); 1$<br>$(0, 1, 0, 2); 1$<br>$(0, 1, 1, 3); 1$<br>$(1, 1, 0, 0); 1$<br>$(1, 1, 1, 1); 1$<br>$(1, 1, 2, 3); 1$<br>$(1, 2, 1, 3); 1$<br>$(1, 2, 2, 4); 1$ | $(0, 0, 1, 0); 1$<br>$(0, 1, 0, 0); 1$<br>$(0, 1, 1, 1); 1$<br>$(0, 1, 2, 3); 1$<br>$(1, 1, 0, 1); 1$<br>$(1, 1, 1, 2); 1$<br>$(1, 2, 0, 2); 1$<br>$(1, 2, 1, 4); 1$<br>$(1, 2, 2, 5); 1$ | $(0, 0, 1, 1); 1$<br>$(0, 1, 0, 1); 1$<br>$(0, 1, 1, 2); 1$<br>$(1, 0, 0, 0); 1$<br>$(1, 1, 0, 2); 1$<br>$(1, 1, 1, 3); 1$<br>$(1, 2, 1, 2); 1$<br>$(1, 2, 2, 3); 1$<br>$(1, 3, 2, 5); 1$<br>$(2, 3, 2, 5); 1$ |

| Example 5                               |                     |                 |                 |
|-----------------------------------------|---------------------|-----------------|-----------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 30, 1, -5)$ |                 |                 |
| $(a_1, \dots, a_N)$                     | $(7, 11, 11)$       |                 |                 |
| (coordinates); multiplicity             | (0, 0, 0, 1); 1     | (0, 0, 1, 0); 1 | (0, 0, 1, 1); 1 |
|                                         | (0, 1, 0, 0); 1     | (0, 1, 0, 1); 1 | (0, 1, 1, 1); 1 |
|                                         | (0, 1, 1, 2); 1     | (1, 0, 0, 0); 1 | (1, 0, 1, 0); 1 |
|                                         | (1, 0, 1, 1); 1     | (1, 1, 1, 1); 1 | (1, 1, 1, 2); 1 |
|                                         | (1, 1, 2, 2); 1     | (1, 1, 2, 3); 1 | (1, 2, 2, 3); 1 |
|                                         | (2, 1, 2, 2); 1     | (2, 1, 2, 3); 1 | (2, 1, 3, 3); 1 |
|                                         | (2, 2, 2, 3); 1     | (2, 2, 3, 3); 1 | (2, 2, 3, 4); 1 |
|                                         | (3, 1, 3, 3); 1     | (3, 2, 3, 3); 1 | (3, 2, 3, 4); 1 |
|                                         | (3, 2, 4, 4); 1     | (3, 2, 4, 5); 1 | (3, 3, 4, 5); 1 |
|                                         | (3, 3, 4, 6); 1     | (3, 3, 5, 6); 1 | (4, 3, 5, 6); 1 |

| Example 6                               |                     |                 |                 |
|-----------------------------------------|---------------------|-----------------|-----------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(3, 1, 32, 1, -7)$ |                 |                 |
| $(a_1, \dots, a_N)$                     | $(7, 11, 13)$       |                 |                 |
| (coordinates); multiplicity             | (0, 0, 0, 1); 1     | (0, 0, 1, 0); 1 | (0, 0, 1, 1); 1 |
|                                         | (0, 0, 1, 2); 1     | (0, 0, 1, 3); 1 | (0, 1, 0, 0); 1 |
|                                         | (0, 1, 0, 1); 1     | (0, 1, 0, 2); 1 | (0, 1, 1, 1); 1 |
|                                         | (0, 1, 1, 2); 1     | (0, 1, 1, 3); 1 | (0, 1, 1, 4); 1 |
|                                         | (0, 1, 2, 4); 1     | (1, 0, 0, 0); 1 | (1, 1, 0, 0); 1 |
|                                         | (1, 1, 0, 1); 1     | (1, 1, 0, 2); 1 | (1, 1, 1, 1); 1 |
|                                         | (1, 1, 1, 2); 1     | (1, 1, 1, 3); 1 | (1, 1, 1, 4); 1 |
|                                         | (1, 1, 2, 4); 1     | (1, 2, 0, 2); 1 | (1, 2, 1, 2); 1 |
|                                         | (1, 2, 1, 3); 1     | (1, 2, 1, 4); 1 | (1, 2, 1, 5); 1 |
|                                         | (1, 2, 2, 4); 1     | (1, 2, 2, 5); 1 | (1, 2, 2, 6); 1 |
|                                         | (1, 3, 2, 6); 1     | (2, 3, 2, 6); 1 |                 |

| Example 7                               |                                                                                                                                                                                              |                                                                                                                                                                                                                      |                                                                                                                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(4, 1, 25, 1, -4)$                                                                                                                                                                          |                                                                                                                                                                                                                      |                                                                                                                                                                                              |
| $(a_1, \dots, a_N)$                     | $(4, 5, 7, 8)$                                                                                                                                                                               |                                                                                                                                                                                                                      |                                                                                                                                                                                              |
| (coordinates); multiplicity             | $(0, 0, 0, 0, 1); 1$<br>$(0, 0, 1, 0, 0); 1$<br>$(0, 0, 1, 1, 1); 1$<br>$(0, 1, 0, 1, 1); 1$<br>$(0, 1, 1, 1, 2); 1$<br>$(1, 0, 0, 1, 1); 1$<br>$(1, 0, 1, 2, 1); 1$<br>$(1, 1, 1, 1, 2); 1$ | $(0, 0, 0, 1, 0); 1$<br>$(0, 0, 1, 0, 1); 1$<br>$(0, 1, 0, 0, 0); 1$<br>$(0, 1, 1, 0, 1); 1$<br>$(1, 0, 0, 0, 0); 1$<br>$(1, 0, 1, 1, 0); 1$<br>$(1, 1, 0, 1, 1); 1$<br>$(1, 1, 1, 2, 1); 1$<br>$(1, 1, 2, 2, 2); 1$ | $(0, 0, 0, 1, 1); 1$<br>$(0, 0, 1, 1, 0); 1$<br>$(0, 1, 0, 0, 1); 1$<br>$(0, 1, 1, 1, 1); 1$<br>$(1, 0, 0, 1, 0); 1$<br>$(1, 0, 1, 1, 1); 1$<br>$(1, 1, 1, 1, 1); 1$<br>$(1, 1, 1, 2, 2); 1$ |

| Example 8                               |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(4, 1, 30, 1, -5)$                                                                                                                                                                                                                          |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                              |
| $(a_1, \dots, a_N)$                     | $(5, 7, 8, 9)$                                                                                                                                                                                                                               |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                              |
| (coordinates); multiplicity             | $(0, 0, 0, 0, 1); 1$<br>$(0, 0, 1, 0, 0); 1$<br>$(0, 0, 1, 1, 1); 1$<br>$(0, 1, 0, 1, 1); 1$<br>$(0, 1, 1, 1, 1); 1$<br>$(1, 0, 0, 0, 0); 1$<br>$(1, 0, 1, 1, 0); 1$<br>$(1, 1, 1, 0, 1); 1$<br>$(1, 1, 1, 2, 2); 1$<br>$(1, 1, 2, 2, 2); 1$ | $(0, 0, 0, 1, 0); 1$<br>$(0, 0, 1, 0, 1); 1$<br>$(0, 1, 0, 0, 0); 1$<br>$(0, 1, 0, 1, 2); 1$<br>$(0, 1, 1, 1, 2); 1$<br>$(1, 0, 1, 0, 0); 1$<br>$(1, 0, 1, 1, 1); 1$<br>$(1, 1, 1, 1, 1); 1$<br>$(1, 1, 2, 1, 1); 1$<br>$(1, 1, 2, 2, 3); 1$ | $(0, 0, 0, 1, 1); 1$<br>$(0, 0, 1, 1, 0); 1$<br>$(0, 1, 0, 0, 1); 1$<br>$(0, 1, 1, 0, 1); 1$<br>$(0, 1, 1, 2, 2); 1$<br>$(1, 0, 1, 0, 1); 1$<br>$(1, 0, 2, 1, 1); 1$<br>$(1, 1, 1, 1, 2); 1$<br>$(1, 1, 2, 1, 2); 1$<br>$(1, 2, 2, 2, 3); 1$ |

| Example 9                               |                     |                    |                    |
|-----------------------------------------|---------------------|--------------------|--------------------|
| $(n, k,  Z , \text{adim}, \text{vdim})$ | $(4, 1, 33, 1, -8)$ |                    |                    |
| $(a_1, \dots, a_N)$                     | $(5, 7, 9, 11)$     |                    |                    |
| (coordinates); multiplicity             | (0, 0, 0, 0, 1); 1  | (0, 0, 0, 1, 0); 1 | (0, 0, 0, 1, 1); 1 |
|                                         | (0, 0, 0, 1, 2); 1  | (0, 0, 1, 0, 0); 1 | (0, 0, 1, 0, 1); 1 |
|                                         | (0, 0, 1, 1, 1); 1  | (0, 0, 1, 1, 2); 1 | (0, 1, 0, 0, 0); 1 |
|                                         | (0, 1, 0, 0, 1); 1  | (0, 1, 0, 1, 1); 1 | (0, 1, 0, 1, 2); 1 |
|                                         | (0, 1, 1, 0, 1); 1  | (0, 1, 1, 1, 1); 1 | (0, 1, 1, 1, 2); 1 |
|                                         | (0, 1, 1, 1, 3); 1  | (0, 1, 1, 2, 3); 1 | (1, 0, 0, 0, 0); 1 |
|                                         | (1, 0, 1, 0, 0); 1  | (1, 0, 1, 0, 1); 1 | (1, 0, 1, 1, 1); 1 |
|                                         | (1, 0, 1, 1, 2); 1  | (1, 0, 2, 1, 2); 1 | (1, 1, 1, 0, 1); 1 |
|                                         | (1, 1, 1, 1, 1); 1  | (1, 1, 1, 1, 2); 1 | (1, 1, 1, 1, 3); 1 |
|                                         | (1, 1, 1, 2, 3); 1  | (1, 1, 2, 1, 2); 1 | (1, 1, 2, 1, 3); 1 |
|                                         | (1, 1, 2, 2, 3); 1  | (1, 1, 2, 2, 4); 1 | (1, 2, 2, 2, 4); 1 |